

No.
Date

$$i \sum_{n=1}^{\infty} \frac{x^n}{2^n n^3} : \lim_{n \rightarrow \infty} = \frac{x^{n+1}}{2^{n+1} (n+1)^3} \cdot \frac{2^n n^3}{x^n}$$

$$= \frac{x^{n+1}}{2^{n+1} (n+1)^3} \cdot \frac{2^n n^3}{x^n}$$

$$= \frac{x}{2} \cdot \frac{n^3}{(n+1)^3} : \frac{n^3}{n^3}$$

$$= \frac{x}{2} \cdot \frac{n^3}{(n+1)^3} : \frac{n^3}{n^3} = \frac{x}{2}$$

$$= \frac{x}{2} < 1$$

$$= x < 2$$

$$-2 < x < 2$$

$$j \sum_{n=1}^{\infty} \frac{x^n}{n!} : \lim_{n \rightarrow \infty} = \frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n}$$

$$= \frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n}$$

$$= \left| \frac{x}{n+1} \right| = 0 \quad \text{maka setiap bilangan real adalah konvergen}$$

Noufal
2340506
Fakultas
Teknologi

2.)

$$\sum_{n=1}^{\infty} n! x^{2n} \quad \lim_{n \rightarrow \infty} \left| \frac{(n+1)! x^{2(n+1)}}{n! x^{2n}} \right|$$

$$= |(n+1) x^2|$$

= 0 maka deret hanya konvergen pada $x=0$

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